

DCIT 208 — Software Engineering

IA1 — Personal System Request-Flow Diagram

Full Name	Antwi Gladys Anobea
Student ID	22376230
Team Name	SEG26-29-LUXBYNESS
Project Title	Salon Booking System
Chosen Feature	Book an Appointment
Date	26 June 2026
Team Repository URL	https://github.com/John-owusu-ansah/DCIT-208-Group-29
Individual Portfolio URL	https://github.com/Dept-of-Comp-Sci-University-of-Ghana/dcit208-individual-engineering-evidence-portfolio-2026-anobea
YouTube Video Link	https://youtube.com/shorts/BHrkF0mwICI?feature=share

AI Disclosure Statement

I used AI assistance. The tool used was Claude (Anthropic). AI was used to help structure the request-flow explanation and suggest system design answers after I had selected my feature and understood the concepts. All outputs were reviewed, adapted to my project, and verified by me. I understand that I must be able to explain and defend every part of this submission.

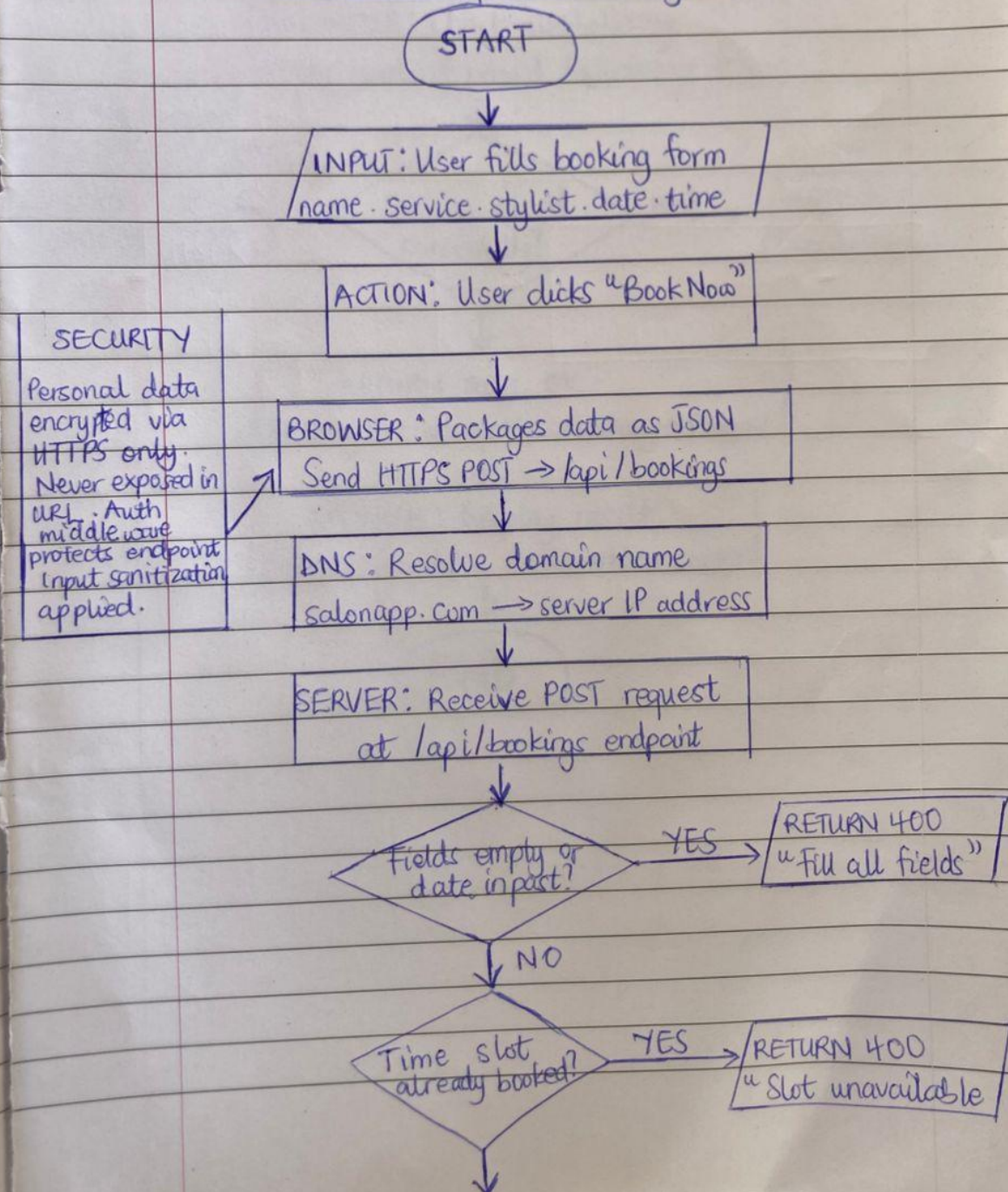
Date	Tool	Prompt Summary	Output Used	Output Rejected	Verification
26 Jun 2026	Claude (Anthropic)	Helped structure request-flow for salon booking system	Flow steps, checklist answers, explanation structure	Raw text was adapted and personalised	Reviewed against lecture notes and assignment brief

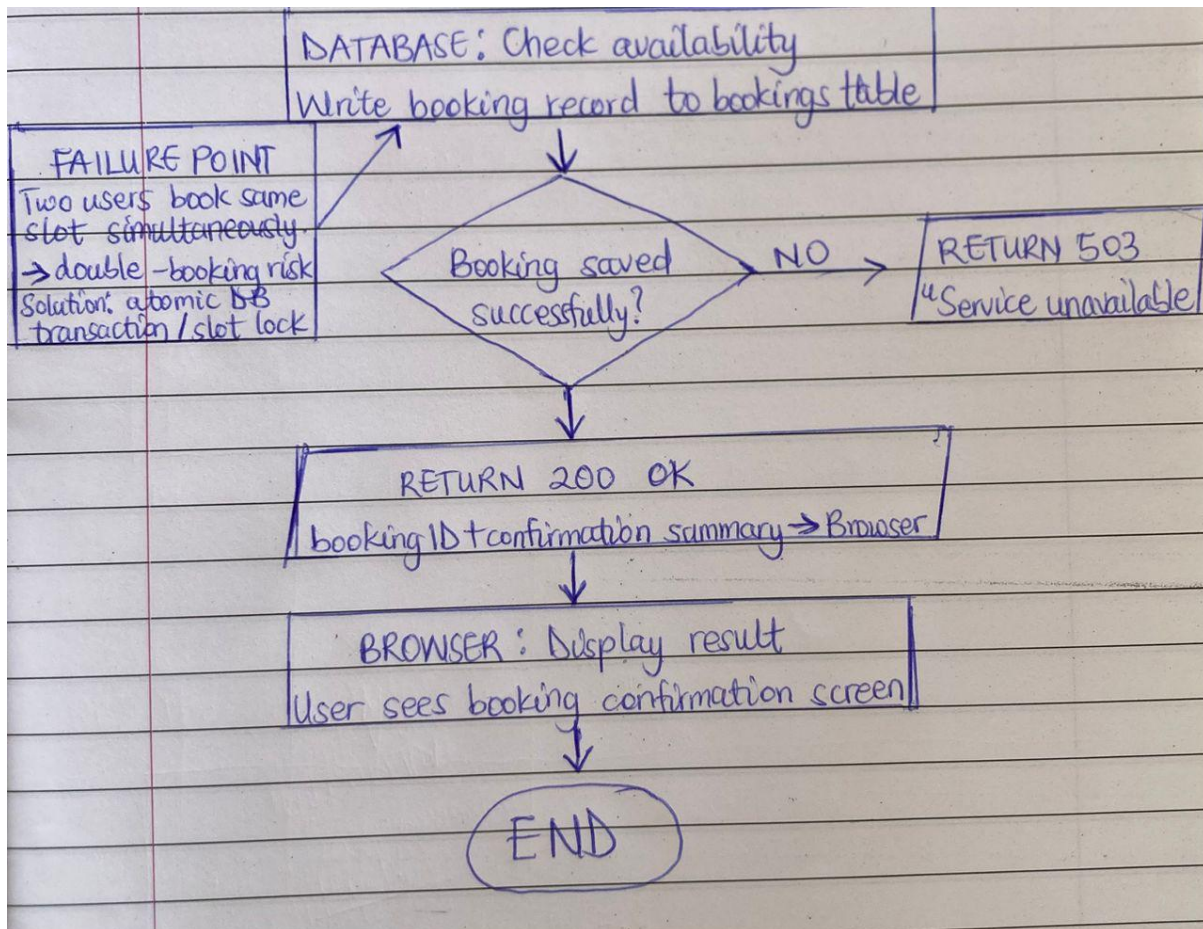
Handwritten Request-Flow Diagram

NAME : Antwi Gladys Anobea
STUDENT ID : 22376230
TEAM : SEG26-29-LUXBYNESS
PROJECT : Salon Booking System
DATE : 25th June, 2026

FEATURE : Book an Appointment

Handwritten Request-Flow Diagram





Typed Explanation: Book an Appointment

Salon Booking System

Introduction

The selected feature for this assignment is the booking functionality of a Salon Booking System. The purpose of this feature is to allow customers to reserve appointments with a salon by selecting a preferred service, date, and time. The request-flow diagram illustrates how a booking request travels from the user's browser to the server, how it is processed, how the database is involved, and how the response is returned to the user. Understanding this flow is important because it shows how different software components work together to provide a reliable and secure booking experience.

1. Device Used by the User

The Salon Booking System is designed as a web application that users can access using various internet-enabled devices such as desktop computers, laptops, tablets, and smartphones. Regardless of the device used, customers interact with the system through a web browser. This provides flexibility because users can make appointments from home or while travelling without needing to install a separate application.

2. Browser or Mobile Application

For this project, the client interface is a web browser. The browser is responsible for displaying the booking form, collecting the customer's input, and communicating with the web server.

After the user completes the booking form and clicks the **Book Now** button, the browser packages the submitted information into an HTTPS request and sends it to the server.

3. URL or Endpoint Involved

When the customer submits the booking form, the browser sends an HTTP POST request to the booking endpoint. Depending on the final implementation, this endpoint may be represented as a route such as /book or /api/bookings. The request contains all the booking information entered by the customer and is sent securely to the application server for processing.

4. Protocol Used

The communication between the browser and the web server uses the HTTPS protocol. HTTPS encrypts all information transmitted between the client and the server. This protects customer information such as names, phone numbers, email addresses, and booking details from being intercepted while travelling across the internet. HTTPS also helps verify that the client is communicating with the legitimate server.

5. Data Sent During the Request

The booking request contains several pieces of information entered by the customer. These may include the customer's full name, phone number, email address, selected salon service, preferred stylist (if available), appointment date, appointment time, and any additional notes. The browser packages this information into the request body before sending it to the server.

6. Request Validation

After receiving the request, the server performs several validation checks before processing the booking. First, it verifies that all required fields have been completed correctly. If any mandatory field is missing or contains invalid data, the server immediately rejects the request and returns a **400 Bad Request** response. This prevents incomplete or incorrect information from reaching the database.

If the required information is valid, the application performs another validation by checking whether the requested appointment slot has already been reserved. This prevents two customers from booking the same stylist or appointment time.

7. Database Interaction

Once validation is successful, the application communicates with the database. The database first checks whether an existing booking already occupies the selected appointment slot. If the slot is available, a new booking record is created and stored. This process ensures that booking information is permanently saved and can later be viewed, modified, or cancelled by authorized users.

The database therefore plays a critical role in maintaining the integrity and reliability of the booking system because it stores all appointment information in an organized manner.

8. Server Response

After processing the request, the server generates an appropriate response.

If the booking is successfully saved, the server returns an **HTTP 200 OK** response together with a confirmation message indicating that the appointment has been booked successfully.

If validation fails because required fields are missing or incorrect, the server returns an **HTTP 400 Bad Request** response.

If the booking cannot be completed because the database or another critical service is temporarily unavailable, the server returns an **HTTP 503 Service Unavailable** response.

The browser receives this response and displays either a confirmation message or an error message to the customer.

9. What Happens if Validation Fails?

Validation is an important part of the booking process because it protects the application from incorrect or incomplete data. If validation fails, the request is rejected immediately. The booking is not stored in the database, and the user receives an error message explaining what needs to be corrected before submitting the form again.

This approach improves system reliability while reducing unnecessary processing by the database.

10. What Happens if the Database is Unavailable?

If the database becomes unavailable due to maintenance, hardware failure, or network problems, the server cannot complete the booking operation. In this situation, the application returns an **HTTP 503 Service Unavailable** response to inform the customer that the service is temporarily unavailable.

No booking is stored during this process, preventing incomplete or inconsistent booking records from being created.

11. What Happens if 500 Users Submit Requests at the Same Time?

A real-world booking system must be capable of handling many users simultaneously. If hundreds of customers attempt to book appointments at the same time, the application server processes multiple requests concurrently. The database should use transactions or locking mechanisms to prevent duplicate bookings for the same appointment slot.

As the system grows, additional techniques such as load balancing, caching, optimized database queries, and horizontal server scaling can be introduced to improve performance and availability during periods of heavy traffic.

12. Security and Privacy Concerns

Protecting customer information is one of the most important requirements of the Salon Booking System. Sensitive information such as customer names, phone numbers, and email addresses should be transmitted using HTTPS encryption.

The application should also validate and sanitize all user input to reduce the risk of SQL injection attacks and other malicious inputs. Authentication should be implemented for salon staff and administrators to ensure that only authorized users can manage bookings. Customer information should be stored securely, and access should be limited to authorized personnel only.

Conclusion

The request-flow diagram demonstrates the complete journey of a booking request within the Salon Booking System. Beginning with the customer's interaction through a web browser, the request passes through DNS resolution, reaches the application server, undergoes validation,

interacts with the database, and finally returns a response to the user. By validating requests before database operations, preventing duplicate bookings, handling system failures appropriately, and implementing security measures, the system provides an efficient, reliable, and secure booking experience. Understanding this request flow helps software engineers design applications that are both user-friendly and capable of supporting real-world business operations.